

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	ITA0517	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)		
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Section / Batch:	B	Date:	21-08-2026

Code of Conduct

/ Yoga Madha A-192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

1. Detection Performance Comparison

Question:

Two object detection models produce the following results:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

Answer:

(a) Precision and Recall Calculation

Precision and recall are important metrics used to evaluate the performance of an object detection model.

Precision measures how many of the objects detected by the model are actually correct.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall measures how many of the actual objects are successfully detected by the model.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Where:

- TP = True Positives
- FP = False Positives
- FN = False Negatives

Model A – Viola-Jones

Given:

- TP = 80
- FP = 20
- FN = 30

Precision:

$$\text{Precision} = 80 / (80 + 20)$$

$$\text{Precision} = 80 / 100$$

$$\text{Precision} = 0.80$$

Therefore,

$$\text{Precision} = 80\%$$

Recall:

$$\text{Recall} = 80 / (80 + 30)$$

$$\text{Recall} = 80 / 110$$

$$\text{Recall} = 0.7273$$

Therefore,

$$\text{Recall} = 72.73\%$$

Thus, Model A has a precision of **80%** and a recall of **72.73%**.

Model B – YOLO

Given:

- TP = 90
- FP = 15
- FN = 20

Precision:

$$\text{Precision} = 90 / (90 + 15)$$

$$\text{Precision} = 90 / 105$$

$$\text{Precision} = 0.8571$$

Therefore,

Precision = 85.71%

Recall:

$\text{Recall} = 90 / (90 + 20)$

$\text{Recall} = 90 / 110$

$\text{Recall} = 0.8182$

Therefore,

Recall = 81.82%

Thus, Model B has a precision of **85.71%** and a recall of **81.82%**.

(b) Comparison and Evaluation

Performance Measure	Model A – Viola-Jones	Model B – YOLO
True Positives	80	90
False Positives	20	15
False Negatives	30	20
Precision	80%	85.71%
Recall	72.73%	81.82%

From the results, YOLO performs better than Viola-Jones in both precision and recall.

The precision of YOLO is **85.71%**, which is higher than the **80%** precision of Viola-Jones.

This means that YOLO produces fewer incorrect positive detections.

The recall of YOLO is **81.82%**, which is higher than the **72.73%** recall of Viola-Jones. This means that YOLO detects a larger proportion of the actual objects in the scene.

YOLO also has fewer false positives, with only **15 false positives**, compared with **20** for Viola-Jones. In addition, YOLO has only **20 false negatives**, compared with **30** for Viola-Jones.

Therefore, YOLO provides better and more reliable object detection performance.

Conclusion:

Model B (YOLO) is the better object detection model. It achieves higher precision and recall while producing fewer false positives and false negatives. Therefore, based on the given results, YOLO is more effective and reliable for object detection.

2. Motion Estimation Error Analysis

Question:

Two motion estimation techniques produce displacement errors (in pixels):

Frame	Method A	Method B
1	2	3
2	3	5
3	2	4

(a) Compute average error for both methods.

(b) Compare and evaluate which method is more accurate.

Answer:

(a) Average Error Calculation

Motion estimation error represents the difference between the estimated movement and the actual movement. A smaller error indicates better and more accurate motion estimation.

The average error is calculated using the formula:

$\text{Average Error} = \text{Sum of Errors} / \text{Number of Frames}$

Method A

The errors for Method A are:

- Frame 1 = 2 pixels

- Frame 2 = 3 pixels
- Frame 3 = 2 pixels

Total error:

$$\text{Total Error} = 2 + 3 + 2$$

$$\text{Total Error} = 7 \text{ pixels}$$

Average error:

$$\text{Average Error} = 7 / 3$$

$$\text{Average Error} = 2.33 \text{ pixels}$$

Therefore,

Average Error of Method A = 2.33 pixels

Method B

The errors for Method B are:

- Frame 1 = 3 pixels
- Frame 2 = 5 pixels
- Frame 3 = 4 pixels

Total error:

$$\text{Total Error} = 3 + 5 + 4$$

$$\text{Total Error} = 12 \text{ pixels}$$

Average error:

$$\text{Average Error} = 12 / 3$$

$$\text{Average Error} = 4.00 \text{ pixels}$$

Therefore,

Average Error of Method B = 4.00 pixels

(b) Comparison and Evaluation

Frame	Method A	Method B
Frame 1	2 px	3 px
Frame 2	3 px	5 px
Frame 3	2 px	4 px
Average Error	2.33 px	4.00 px

Method A has an average error of **2.33 pixels**, while Method B has an average error of **4.00 pixels**.

The difference between the average errors is:

$$\text{Difference} = 4.00 - 2.33$$

$$\text{Difference} = 1.67 \text{ pixels}$$

Therefore, Method A has an average error that is **1.67 pixels lower** than Method B.

Method A also performs better in every individual frame. In Frame 1, Method A has an error of 2 pixels compared with 3 pixels for Method B. In Frame 2, Method A has an error of 3 pixels compared with 5 pixels for Method B. In Frame 3, Method A has an error of 2 pixels compared with 4 pixels for Method B.

Since a lower displacement error indicates more accurate motion estimation, Method A is more accurate.

Conclusion:

Method A is the more accurate motion estimation technique. Its average error is only **2.33 pixels**, compared with **4.00 pixels** for Method B. Therefore, Method A provides more precise and consistent motion estimation.

3. Optical Flow Magnitude Comparison

Question:

Optical flow magnitudes are given below in pixels/frame:

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

(a) Compute average motion magnitude for both methods.

(b) Compare and evaluate which method captures stronger motion.

Answer:

(a) Average Motion Magnitude Calculation

Optical flow is used to estimate the apparent movement of objects or pixels between consecutive frames. The optical flow magnitude represents the amount of motion detected in pixels per frame.

The average motion magnitude is calculated using:

Average Motion Magnitude = Sum of Motion Magnitudes / Number of Regions

Method A

The motion magnitudes for Method A are:

- R1 = 5 pixels/frame
- R2 = 6 pixels/frame
- R3 = 5 pixels/frame

Total motion magnitude:

Total = 5 + 6 + 5

Total = 16 pixels/frame

Average motion magnitude:

Average = 16 / 3

Average = 5.33 pixels/frame

Therefore,

Average Motion Magnitude of Method A = 5.33 pixels/frame

Method B

The motion magnitudes for Method B are:

- R1 = 8 pixels/frame
- R2 = 10 pixels/frame
- R3 = 9 pixels/frame

Total motion magnitude:

Total = 8 + 10 + 9

Total = 27 pixels/frame

Average motion magnitude:

Average = 27 / 3

Average = 9.00 pixels/frame

Therefore,

Average Motion Magnitude of Method B = 9.00 pixels/frame

(b) Comparison and Evaluation

Region	Method A	Method B
R1	5 px/frame	8 px/frame
R2	6 px/frame	10 px/frame
R3	5 px/frame	9 px/frame

Region	Method A	Method B
Average	5.33 px/frame	9.00 px/frame

Method B has an average motion magnitude of **9.00 pixels/frame**, while Method A has an average motion magnitude of **5.33 pixels/frame**.

The difference between the average motion magnitudes is:

Difference = $9.00 - 5.33$

Difference = 3.67 pixels/frame

Therefore, Method B captures an average motion magnitude that is **3.67 pixels/frame higher** than Method A.

Method B also produces a higher motion magnitude in every region. In R1, Method B detects 8 pixels/frame compared with 5 pixels/frame for Method A. In R2, Method B detects 10 pixels/frame compared with 6 pixels/frame. In R3, Method B detects 9 pixels/frame compared with 5 pixels/frame.

A higher optical flow magnitude indicates stronger apparent movement. Therefore, Method B captures stronger motion across all three regions.

However, a higher motion magnitude does not automatically mean that Method B is more accurate. To determine accuracy, actual ground-truth motion would be required. In this question, the requirement is specifically to identify which method captures **stronger motion**, so the method with the higher average magnitude is selected.

Conclusion:

Method B captures stronger motion. It has an average optical flow magnitude of **9.00 pixels/frame**, compared with **5.33 pixels/frame** for Method A. Therefore, Method B detects greater apparent movement across the observed regions.